D03 · Shop Floor Execution & MES

02 · Work order execution & operator interfaces

Supply Chain Reference Architecture v2.2 · mrsupplychain.ai

1 WHAT IT IS

Work order execution is the chain of events between the MES releasing a job and the operator confirming it complete — electronic work instructions, material kit verification, real-time confirmation, and the operator UX that determines whether process discipline holds. It is the most operator-visible layer of the entire manufacturing stack; the failure modes are usually not technical but UX-driven.

2 WHY IT MATTERS

The work order is the smallest unit of execution discipline in a manufacturing plant — and the place where MES theory meets shop floor reality. **Every operator hour spent navigating a poorly-designed work order interface is an hour subtracted from value-adding production**. The cost is asymmetric: a 2-minute delay per work order in a plant doing 800 work orders per shift is 26 hours of lost productive labor per day. The dominant failure pattern is treating the work order as a transactional event (start, confirm, close) rather than an information channel — operators need to see the right work instruction at the right step, the right material kit confirmed, the right quality checks prompted, all while focused on the physical task **(Liker, 2004 — Toyota Production System scholarship)**.

3 HOW IT IS DONE

1. **Eliminate paper traveler entirely — or do not start.** Hybrid paper-electronic execution is worse than pure paper because operators stop trusting the electronic system. The transition must be clean: from day one, the operator sees only the screen, and the paper goes away.
2. **Design the work instruction at the operator station, not the engineering office.** Engineering writes work instructions for engineers. The operator UX must be redesigned by people who watch operators do the task — fewer words, more visuals, context-aware (the right step shown at the right moment based on operator action).
3. **Confirm material kit before the work order starts.** Scanning each material kit against the BOM before production begins eliminates the worst defect category — building with the wrong component. This single discipline prevents an entire class of quality escapes.
4. **Trigger quality checks contextually within the work order flow.** A quality check listed in a separate document is a quality check that does not happen. The MES must prompt the operator at the exact step where the check applies, with the acceptance criteria visible and the result captured before the next step unlocks.
5. **Confirm completion at the cell, not at end-of-shift.** Real-time confirmation creates real-time visibility. Batch confirmation at end-of-shift creates the lag that hides the problem the supervisor needed to know about three hours ago. The operator UX must make confirmation a one-tap action.
6. **Instrument the operator interface and iterate.** Track time-per-step, abandonment, error rates. The operator UX is a product, not a deliverable — the v1 is wrong, the v2 is better, the v3 is the one that works. Without telemetry on operator interaction, the UX never improves.

> **WORKED EXAMPLE — ILLUSTRATIVE**
>
> An electronics assembly plant running 32 lines and 1,400 work orders per day implements electronic work instructions on touch-screen terminals at each cell. Baseline: average time per work order start = 4.2 minutes, of which 2.8 minutes is paperwork. New target: <90 seconds total. The implementation includes material kit barcode scanning, context-aware work instructions, and one-tap completion. Results at 90 days: **average work order start time = 67 seconds, defect-from-wrong-component down 84%, end-of-shift overtime for paperwork closure eliminated entirely**. Saved labor hours: **~3,100 per month across all lines**; the avoided defects (re-work + scrap + warranty) are worth another $180K monthly.

IN PRACTICE

Operations owns the operator UX, not IT alone

The MES vendor's default screens are designed for demos, not for operators who have done 600 cycles before lunch. A small operations-owned team — often called "shop floor experience" — must take continuous ownership of the screens.

Iterate monthly from telemetry and operator feedback

Click counts, dwell time per step, abandonment of optional fields — the same metrics consumer products use. A monthly review cycle with one or two screen changes per cell compounds into 10–20% recovered operator time within a year.

Refuse vendor default screens as the production reality

The demo screens that won the deal are not the screens that should run the floor. Building a configuration backlog from operator complaints in the first 30 days post-go-live is the cheapest improvement available.

Eliminate paper completely as a non-negotiable

Paper work instructions defeat every traceability, version-control, and audit-trail benefit of having an MES **(Liker, 2004 — Toyota Production System scholarship)**. The plants that retain paper as a "backup" have neither system — they have two parallel systems that disagree.

WHAT YOU WOULD USE

OPERATOR EXECUTION PLATFORMS

Tulip — Low-code operator-facing platform designed for rapid iteration of work instructions, with strong app-building tools that let operations teams own the UX without writing code **(Tulip — vendor)**.

VKS (Visual Knowledge Share) — Specialized in visual work instructions for discrete assembly, particularly strong in mixed-model lines and high-variety operations **(VKS — vendor)**.

Native MES operator portals (Siemens, Rockwell, AVEVA) — The standard option when the MES is already in place — strongest when configured by an operations-led team, weakest when left as the vendor default **(Siemens — vendor; Rockwell — vendor; AVEVA — vendor)**.

Custom React/Angular operator apps on top of MES APIs — For operations with the in-house capability — the highest ceiling of UX quality, the most ongoing maintenance commitment. A path, not a destination.

4 SO WHAT

The work order interface is the highest-frequency UX in the operation, and it deserves the discipline of a consumer product, not the indulgence of an enterprise tool. Eliminate paper completely, design the screens at the operator station, confirm material kits before production starts, prompt quality checks contextually, capture completion in real time, and iterate the UX monthly from telemetry. The plants that get this right buy back operator hours measured in thousands per month — at zero additional capex.

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